



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

31-05-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 31-05-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C	A,B	A,C,D	B,C	B,C	A,C	A,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	B	B	B	D	A	
SEC-3	Q.	1	2	3	4	5		
	A.	8	2	3	3	3		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C,D	B	A,D	B,C	A,D	A,B,D	A,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	D	B	A	B	D	
SEC-3	Q.	1	2	3	4	5		
	A.	4	5	3	9	4		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C,D	D	A,B	A,B,D	A,C,D	A,C	B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	B	B	D	D	C	
SEC-3	Q.	1	2	3	4	5		
	A.	2	4	6	1	3		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A,B,C)

$$\frac{Q_{out}}{Q_{in}} = \frac{T_{out}}{T_{in}} \text{ and } Q_{out} = Q_{in} + W$$

$$\Rightarrow Q_{in} = W \left(\frac{T_L}{T_H - T_L} \right)$$

$$\frac{Q_{in}}{t} = P \left(\frac{T_L}{T_H - T_L} \right)$$

2. Ans (A,B)

Between faces LMNO and PQRS

$$dR = \frac{\rho dr}{t \frac{\pi}{2} r}$$

$$R = \frac{2\rho}{t\pi} \ln \frac{b}{a}$$

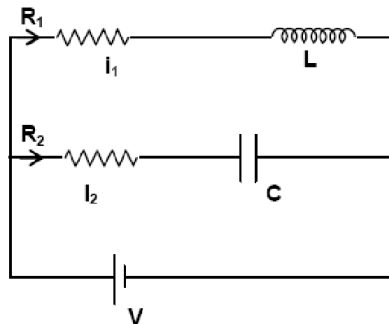
Between faces LMRS and ONQP for an element at distance r from O.

$$dR = \frac{\rho \frac{\pi}{2} r}{t dr}$$

$$\text{So, } \frac{1}{R} = \int \frac{1}{dR} = \int_a^b \frac{t dr}{\rho \frac{\pi}{2} r} = \frac{2t}{\rho\pi} \ln \left(\frac{b}{a} \right)$$

$$R = \frac{\rho\pi}{2t \ln \left(\frac{b}{a} \right)}$$

3. Ans (A, C, D)



$$(i_1)_0 = \frac{50}{25} = 2 \text{ amp}$$

$$(i_2)_0 = \frac{50}{5000} = 0.01 \text{ amp}$$

$$\tau_1 = 0.125 \text{ sec}$$

$$\tau_2 = 0.125 \text{ sec}$$

$$i_2 = 200 i_1$$

$$400 \left(1 - e^{-\frac{t}{0.125}}\right) = \left(0.01 e^{-\frac{t}{0.125}}\right)$$

$$4 \left(\frac{q}{C}\right) = L \frac{di_1}{dt}$$

$$4 \times 50 \left(1 - e^{-\frac{t}{0.125}}\right) = 50 e^{-\frac{t}{0.125}}$$

$$t = \frac{1}{8} \ln \left(\frac{5}{4}\right)$$

4. Ans (B, C)

$$x^2 = \frac{2a}{\sqrt{3}} y$$

$$2x = \frac{2}{\sqrt{3}} a \frac{dy}{dx}$$

$$\tan \theta = \sqrt{3} \frac{x}{a}$$

$$\theta = 60^\circ$$

$$a = \frac{g \sin \theta}{1 + \frac{I_{cm}}{mR^2}} = \frac{g}{\sqrt{3}}$$

$$f = \frac{mg \sin \theta}{1 + \frac{mR^2}{I_{cm}}} = \frac{mg}{2\sqrt{3}}$$

6. Ans (A, C)

$$v_0 = 4\sqrt{gR}$$

$$\vec{J} = \Delta \vec{P}$$

$$-\int T dt = 2M (v - 4\sqrt{gR})$$

$$-\int T dt = 2M (R\omega - 4\sqrt{gR}) \quad \dots (i)$$

$$\left(\int T dt\right) R = MR^2 \omega \quad \dots (ii)$$

From (i) and (ii)

$$\omega = \frac{8}{3} \sqrt{\frac{g}{R}} \text{ and } v = \frac{8}{3} \sqrt{gR}$$

7. Ans (A, C)

$$x_{cm} = \frac{\int x dm}{\int dm} = \frac{\int_0^L x \lambda_0 \left(1 + \frac{x}{L}\right) dx}{\int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) dx} = \frac{5L}{9}$$

$$m = \int dm = \frac{3\lambda_0 L}{2}$$

$$\left[\left(A \cdot \frac{L}{9}\right) \rho g\right] \frac{17}{18} L \sin \theta = F \cdot L \cos \theta$$

PHYSICS

SEC-2

1. Ans (A)

$$(I) \Delta x = d \sin \theta = 1 \times 0.2 = 0.2 \text{ mm} = (400)\lambda$$

$$\text{hence intensity} = 4I_0$$

$$(II) \Delta x = 0 \text{ hence}$$

$$I = (\sqrt{I_0} + \sqrt{2I_0})^2 = 5.8I_0$$

$$(III) \Delta x (\mu - 1)t = 0.25 \mu\text{m}$$

$$\text{also } \Delta x = \frac{\lambda}{2} \text{ hence } I = 0$$

$$(IV) \Delta x = (\mu_1 - \mu_2)t = \frac{1}{6} \mu\text{m}$$

$$\text{also } \Delta x = \frac{\lambda}{3} \Rightarrow \Delta \phi = 120^\circ$$

$$I = I_0 + I_0 + 2\sqrt{I_0} \sqrt{I_0} \left(-\frac{1}{2}\right) = I_0$$

2. Ans (B)

$$(I) \Delta x = d \sin \theta = 1 \times 0.2 = 0.2 \text{ mm} = (400)\lambda$$

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4. Ans (B)

$$\frac{1}{2}ma_{\text{cm}}^2t^2 + \frac{1}{2}I_{\text{cm}}\alpha^2t^2$$

In each configuration kinetic energy of solid sphere will be maximum.

When $F_{\text{net}} = 3F$ the kinetic energy of sphere will be maximum.

$$= \frac{14F^2}{m}t$$

When $F_1 = F_2 = F$

$R\alpha$ $\uparrow$			
sphere	cylinder	Spherical shell	Cylindrical shell
$\frac{5F}{2m}$	$\frac{2F}{m}$	$\frac{3F}{2m}$	$\frac{F}{m}$
$\frac{5F}{m}$	$\frac{4F}{m}$	$\frac{3F}{m}$	$\frac{2F}{m}$
$\frac{5F}{2m}$	$\frac{2F}{m}$	$\frac{3F}{2m}$	$\frac{F}{m}$
$\frac{5F}{2m}$	$\frac{2F}{m}$	$\frac{3F}{2m}$	$\frac{2F}{m}$

$$a_{\text{cm}} = \frac{2F}{m}, R\alpha = \frac{FR^2}{I_{\text{cm}}}$$

When $F_2 = 2F_1 = 2F$

$$a_{\text{cm}} = \frac{3F}{m}, R\alpha = \frac{2FR^2}{I_{\text{cm}}}$$

When $F_1 = 0$ and $F_2 = F$

$$a_{\text{cm}} = \frac{F}{m}, R\alpha = \frac{FR^2}{I_{\text{cm}}}$$

$$\text{When } F_1 = \frac{3F_2}{2} = \frac{3F}{2}$$

$$a_{\text{cm}} = \frac{5F}{2m}, R\alpha = \frac{FR^2}{I_{\text{cm}}}$$

5. Ans (D)

$$\frac{1}{2}ma_{\text{cm}}^2t^2 + \frac{1}{2}I_{\text{cm}}\alpha^2t^2$$

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$$a_{\text{cm}} = \frac{3F}{m}, R\alpha = \frac{2FR^2}{I_{\text{cm}}}$$

When $F_1 = 0$ and $F_2 = F$

$$a_{\text{cm}} = \frac{F}{m}, R\alpha = \frac{FR^2}{I_{\text{cm}}}$$

$$\text{When } F_1 = \frac{3F_2}{2} = \frac{3F}{2}$$

$$a_{\text{cm}} = \frac{5F}{2m}, R\alpha = \frac{FR^2}{I_{\text{cm}}}$$

6. Ans (A)

$$\frac{1}{2} m a_{cm}^2 t^2 + \frac{1}{2} I_{cm} \alpha^2 t^2$$

In each configuration kinetic energy of solid sphere will be maximum.

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Rα			
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$\frac{5F}{m}$	$\frac{4F}{m}$	$\frac{3F}{m}$	$\frac{2F}{m}$
$\frac{5F}{2m}$	$\frac{2F}{m}$	$\frac{3F}{2m}$	$\frac{F}{m}$
$\frac{5F}{2m}$	$\frac{2F}{m}$	$\frac{3F}{2m}$	$\frac{2F}{m}$

$$a_{cm} = \frac{2F}{m}, R\alpha = \frac{FR^2}{I_{cm}}$$

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$$a_{cm} = \frac{3F}{m}, R\alpha = \frac{2FR^2}{I_{cm}}$$

When $F_1 = 0$ and $F_2 = F$

$$a_{cm} = \frac{F}{m}, R\alpha = \frac{FR^2}{I_{cm}}$$

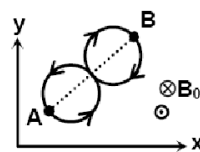
$$\text{When } F_1 = \frac{3F_2}{2} = \frac{3F}{2}$$

$$a_{cm} = \frac{5F}{2m}, R\alpha = \frac{FR^2}{I_{cm}}$$

PHYSICS

SEC-3

1. Ans (8)



$$AB = \frac{4mv}{qB_0} = 8 \text{ metre}$$

2. Ans (2)

$$e \times \frac{kQ}{R} = 1.44 \text{ keV} \Rightarrow \text{charge is constant}$$

$$\Rightarrow \frac{kQ}{R} = 1440 \text{ V}$$

$$\Rightarrow Q = \frac{1440}{9 \times 10^9} \text{ V} = Ne$$

$$N = \frac{1440}{9 \times 10^9 \times 1.6} \times 10^{19} = 10^{12}$$

$$N = N_0 (1 - e^{-\lambda t})$$

$$10^{12} = 4 \times \frac{4}{3} \times 10^{12} (1 - e^{-\lambda t})$$

$$\Rightarrow e^{-\lambda t} = \frac{1}{4} \Rightarrow 2 \text{ half lives} \Rightarrow 2 \text{ hrs}$$

$$N_d = 10^{12}$$

$$N_s = \frac{1}{3} \times 10^{12} = \frac{N_0}{4}$$

$$\Rightarrow 2 \text{ half lives.}$$

3. Ans (3)

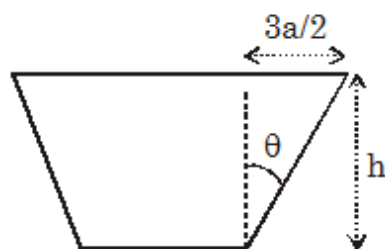
$$C = \frac{2\pi ab\epsilon_0 (k+1)}{(b-a)}$$

$$R = \frac{1}{\alpha} \frac{(b-a)}{\pi ab}$$

$$RC = \frac{2\pi ab\epsilon_0 (k+1)}{(b-a)} \frac{(b-a)}{2\pi ab}$$

$$Q = Q_0 e^{-\frac{+Q}{\epsilon_0(k+1)}}$$

4. Ans (3)



$$\tan \theta = \frac{3a}{2h}$$

$$2 \times 4a T \times \cos \theta = 4a \lambda g$$

$$\cos \theta = \frac{\lambda g}{2T}$$

$$\Rightarrow h = \frac{3\lambda ga}{2\sqrt{4T^2 - \lambda^2 g^2}}$$

5. Ans (3)

$$\frac{v}{4\ell} = 412.5 \text{ (3rd harmonic)}$$

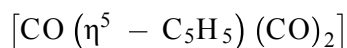
CHEMISTRY

SEC-1

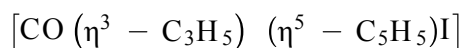
2. Ans (B)

6 geometrical isomers (Case I) + 40 optical isomers (Case II) = 46 space isomers.

3. Ans (A,D)



$$\text{EAN} = 27 - 1 + 10 = 36$$



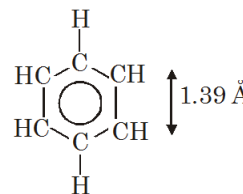
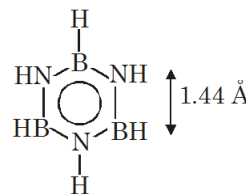
$$\text{EAN} = 27 - 3 + 12 = 36$$

II has no plane of symmetry hence it is optically active

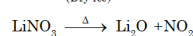
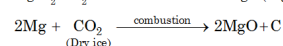
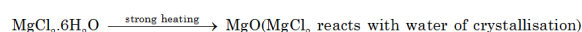
CO act as a catalytic poison in Haber's process.

4. Ans (B,C)

Organic benzene being a more symmetrical molecule has higher melting point (6°C) than inorganic benzene (melting point = -58°C).



6. Ans (A,B,D)



CHEMISTRY

SEC-2

6. Ans (D)

Peptisation : Phenomenon in which a precipitate is passed into colloidal form by adsorption of common ions on its surface.

Delta formation : When river water (sol) meets sea water (electrolyte) coagulation occurs and delta is formed.

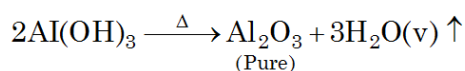
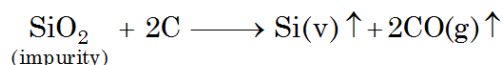
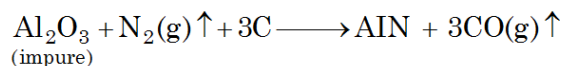
Tyndall effect : Scattering of light rays by suspended particles in a colloid is called Tyndall effect and sky looks blue due Tyndall effect.

Protection of colloids : Protective power of colloids is inversely related to Gold number.

CHEMISTRY

SEC-3

2. Ans (5)



4. Ans (9)

$$[Al^{+3}][OH^-]^3 = 1 \times 10^{-20}$$

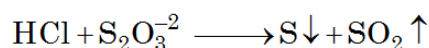
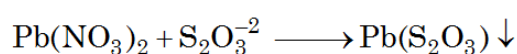
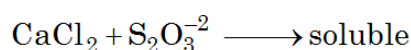
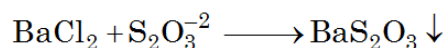
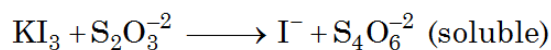
$$10^{-5}[OH^-]^3 = 1 \times 10^{-20}$$

$$[OH^-] = 10^{-5}$$

$$pOH = 5$$

$$pH = 9$$

5. Ans (4)



MATHEMATICS

SEC-1

1. Ans (A,B,C,D)

$$P = \sum_{n=1}^{10} n. \sum_{r=1}^{10} \frac{r^2}{r+n}$$

$$r \leftrightarrow n$$

$$P = \sum_{r=1}^{10} \sum_{n=1}^{10} r. \frac{n^2}{n+r}$$

$$2P = \sum_{r=1}^{10} \sum_{n=1}^{10} \frac{r \cdot n}{(n+r)} (r+n)$$

$$2P = \left(\frac{10 \times 11}{2} \right)^2$$

2. Ans (D)

Function is continuous and differentiable at $x = 0$ and discontinuous everywhere else

3. Ans (A,B)

$$BC = \begin{bmatrix} P_{11} & -1 \\ P_{12} & P_{22} \end{bmatrix}$$

$$ABC = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -a^2 & a+2 \end{bmatrix} \begin{bmatrix} p_{11} & -1 \\ p_{12} & p_{22} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

$$\text{by solving } a = 0, a = -\frac{3}{2}$$

4. Ans (A,B,D)

$y = x + 1$ is parallel to axis of the parabola so reflected ray will pass through its focus (2,2)

5. Ans (A,C,D)

Lines L_1 and L_2 intersect at point (1, 0, 2), L_1, L_2 and $\vec{r} = 4\hat{j} + \lambda(\hat{i} + 2\hat{k})$ are always coplanar $\forall \lambda \in \mathbb{R}$.

Any line passing through point A(1, μ, 3) and intersecting L_1 and L_2 will always pass through (1, 0, 2)

6. Ans (A,C)

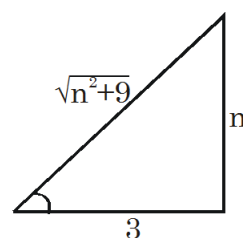
$$\frac{e^x + e^{4/x}}{x} > e^x \text{ for } x \in (1, 2) \text{ and for option}$$

(A)

$$\text{put } x = \frac{4}{t}$$

7. Ans (B,D)

$$T_r = \tan^{-1} \left(\frac{\frac{1}{3}}{1 + \frac{r(r-1)}{9}} \right) = \tan^{-1} \left(\frac{r}{3} \right) - \tan^{-1} \left(\frac{r-1}{3} \right)$$



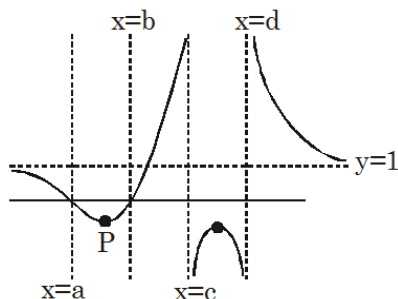
$$S_n = \sum_{r=1}^n T_r = \tan^{-1} \left(\frac{n}{3} \right)$$

MATHEMATICS

SEC-2

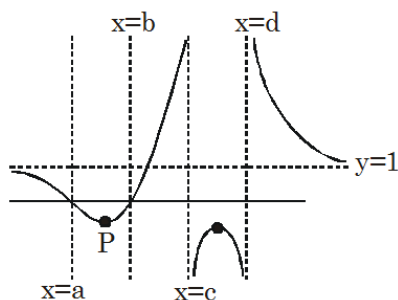
1. **Ans (B)**

We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



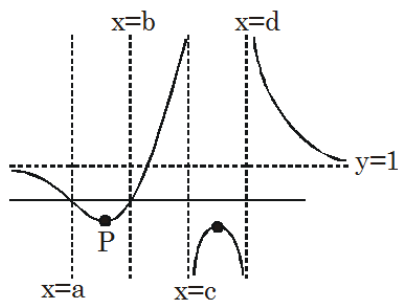
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We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



3. **Ans (B)**

We cannot be sure about the coordinates of P and Q as they depend on a, b, c and d



4. **Ans (D)**

$$\begin{aligned} \int_0^x \sqrt{1 - (f''(t))^2} dt &= \int_0^x f'(t) dt \Rightarrow \sqrt{1 - (f''(x))^2} \\ &= f'(x) \Rightarrow f''(x) = \pm \sqrt{1 - (f'(x))^2} \\ &\Rightarrow x = \pm \sin^{-1} f'(x) \Rightarrow f(x) = \pm (1 - \cos x) \\ &= \pm 2 \sin^2 \frac{x}{2} \\ f(x) &= 2 \sin^2 \frac{x}{2} \text{ or } f(x) = -2 \sin^2 \frac{x}{2} \text{ (reject)} \\ f(x) &\leq \frac{x^2}{2}, \forall x \in [0, 1] \end{aligned}$$

5. **Ans (D)**

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MATHEMATICS

SEC-3

1. **Ans (2)**

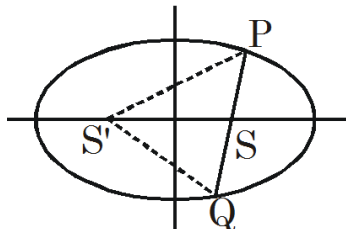
$y = \sin 2x$ and $y = \sin^2 x + \sin x - 1$ will intersect twice in $(0, 2\pi)$

2. **Ans (4)**

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$

$$\Rightarrow \frac{z_1 \bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1 \bar{z}_2}{|z_1 z_2|} = 2 \cos(\theta_1 - \theta_2) + 2 \cos(\theta_1 + \theta_2)$$

3. **Ans (6)**



$$\frac{(S'P + SP) + (SQ + S'Q)}{2} = 2a$$

4. **Ans (1)**

Given $a + b + c = 1$

$9a + 3b + c = 7$

$18 < 25a + 5b + c < 22$

$\Rightarrow$ From above (1), (2) and (3)

$4 < 7a - b - c < 8$

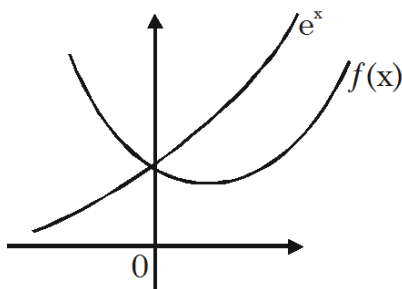
$4 < 7a - b + a + b - 1 < 8$

$5 < 8a < 9$

$\frac{5}{8} < a < \frac{9}{8} \quad (a = 1)$

$a = 1, b = -1, c = 1$

$f(x) = x^2 - x + 1$



5. **Ans (3)**

$$\lim_{x \rightarrow \infty} \frac{e^2 \left[e^{x \ln \left(\frac{x+1}{x-1} \right) - 2} - 1 \right] \left(x \ln \left(\frac{x+1}{x-1} \right) - 2 \right)}{\left(\frac{1}{x^2} \right) \left(x \ln \left(\frac{x+1}{x-1} \right) - 2 \right)}$$

$L = \frac{2}{3} e^2$